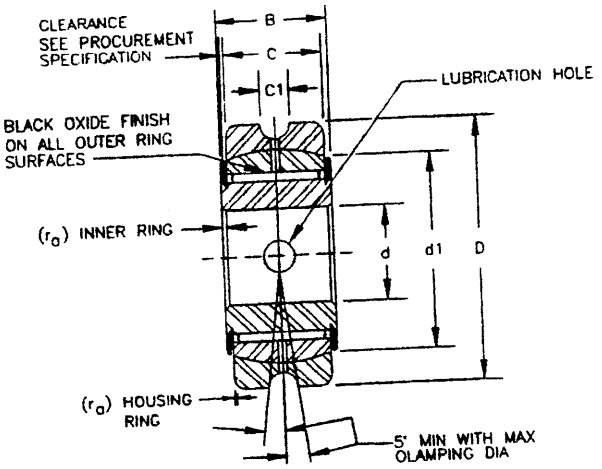
THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-B-3990

THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

Form Approved
OMB No. 0704-0188

INCH DESIGN



DIMENSIONS IN INCHES

DASH NO.	ød BORE	D HOUSING RING OUTSIDE DIA.	D OVERALL WIDTH	C OUTER RING WIDTH	d1 WASHER OUTSIDE DIA	C1 LUBRICATION GROOVE WIDTH	ra 1/ FILLET MAX	TOTAL RADIAL PLAY MAX	da CLAMPING DIAMETER MAX	da CLAMPING DIAMETER MIN	LIMIT LOAD RATING LBF	MASS (APPROX) LB	HOUSING SIZE GAGE +0.0000 −0.0001 LOW	HOUSING SIZE GAGE +0.0000 −0.0001 HIGH
-3	0.1900	0.8750	0.312	0.218	0.625	0.042	0.022	0.0029	0.525	0.438	1800	0.041	0.8742	0.8747
-4	0.2500	0.9375	0.375	0.281	0.687	0.093	0.022	0.0029	0.688	0.516	2870	0.053	0.9367	0.9372
-5	0.3125	1.0625	0.437	0.344	0.750	0.093	0.022	0.0029	0.734	0.578	4070	0.079	1.0617	1.0622

DIMENSIONS IN MILLIMETERS

DASH NO.	ød BORE	D HOUSING RING OUTSIDE DIA.	D OVERALL WIDTH	C OUTER RING WIDTH	d1 WASHER OUTSIDE DIA	C1 LUBRICATION GROOVE WIDTH	ra 1/ FILLET MAX	TOTAL RADIAL PLAY MAX	da CLAMPING DIAMETER MAX	da CLAMPING DIAMETER MIN	LIMIT LOAD RATING M	MASS (APPROX) KG	HOUSING SIZE GAGE +0.0000 −0.0003 LOW	HOUSING SIZE GAGE +0.0000 −0.0003 HIGH
-3	4.826	22.225	7.92	5.54	15.88	1.37	0.6	0.074	15.88	11.13	8000	0.019	22.205	22.217
-4	6.350	22.812	9.52	7.14	17.45	2.36	0.6	0.074	17.48	13.11	12700	0.024	23.792	23.605
-5	7.938	26.988	11.10	8.74	19.05	2.36	0.6	0.074	18.64	14.68	18100	0.036	26.967	26.950

1/ THE CHAMFER ON BEARINGS MUST CLEAR THE MAXIMUM FILLET RADIUS GIVEN IN THE TABLE. THIS SPECIFICATION DOES NOT CONTROL BEARING CHAMFER CONTOURS.

(H) DENOTES CHANGE

INCH-POUND

PREPARING ACTIVITY: NAVY-AS
CUSTODIANS: ARMY- AT NAVY- AS
AIR FORCE- 99 DLA-
REVIEW AF-84 DLA-IS
USER
PROJECT NUMBER: 5110-0856

MILITARY SPECIFICATION SHEET

TITLE
BEARING, ROLLER, NEEDLE-SINGLE ROW, HEAVY DUTY, SELF-ALIGNING, TYPE III, ANTIFRICTION, INCH

SPECIFICATION SHEET NUMBER
MS24463
05 DEC 94
REV H

SUPERSEDING
MS24463G 17 JUN 77

AMSC- N/A FSC 3110

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TABLE I-TOLERANCE LIMITS

DIMENSIONS IN INCHES

⌀d BASIC BORE		ALLOWABLE DEVIATION FROM d OF SINGLE MEAN DIA, d		ALLOWABLE DEVIATION FROM OVER-ALL WIDTH, B		ALLOWABLE DEVIATION FROM WASHER OUTSIDE DIA, d_1	
OVER	INCL	HIGH	LOW	HIGH	LOW	HIGH	LOW
0.1250	0.3125	+0	-0.0007	+0	-0.005	+0.010	-0.010

⌀D BASIC OUTSIDE DIAMETER		ALLOWABLE DEVIATION FROM D OF SINGLE MASS DIA., D		ALLOWABLE DEVIATION FROM OUTER RING WIDTH, C		ALLOWABLE DEVIATION FROM LUBRICATION GROOVE WIDTH, C_1	
OVER	INCL	HIGH	LOW	HIGH	LOW	HIGH	LOW
0.7500	1.0625	+0	-0.0005	+0	-0.005	+0.031	-0.031

DIMENSIONS IN MILLIMETERS

⌀d BASIC BORE		ALLOWABLE DEVIATION FROM d OF SINGLE MEAN DIA, d		ALLOWABLE DEVIATION FROM OVER-ALL WIDTH, B		ALLOWABLE DEVIATION FROM WASHER OUTSIDE DIA, d_1	
OVER	INCL	HIGH	LOW	HIGH	LOW	HIGH	LOW
3.175	7.938	+0	-0.0018	+0	-0.13	+0.25	-0.25

⌀D BASIC OUTSIDE DIAMETER		ALLOWABLE DEVIATION FROM D OF SINGLE MASS DIA., D		ALLOWABLE DEVIATION FROM OUTER RING WIDTH, C		ALLOWABLE DEVIATION FROM LUBRICATION GROOVE WIDTH, C_1	
OVER	INCL	HIGH	LOW	HIGH	LOW	HIGH	LOW
19.050	26.988	+0	-0.013	+0	-0.13	+0.80	-0.80

TABLE II OIL HOLE DATA

BORE-DASH NO.		NUMBER OF HOLES		
OVER	INCL	INNER RING	OUTER RING	SPHERICAL HOUSING
2	5	NONE	2	2

REQUIREMENTS:

1. MATERIAL STEEL, MIL-S-8690, MIL-S-7493, QQ-S-624, QQ-S-631, QQ-S-700, FED STD NO. 66 AISI/SAE STEEL NO. 50100, 51100, 52100.

(H) 2. PLATING. ZINC-NICKEL IN ACCORDANCE WITH AMS 2417, TYPE 2, OR CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2, WITH A MINIMUM THICKNESS OF .0003 INCHES, AND MAXIMUM THICKNESS OF .0006 INCHES.

3. MACHINE FINISH ANSI/ASME B46.1, SEE PRECUREMENT SPECIFICATION.

4. LUBRICATION: GREASE CONFORMING TO MIL-G-81322

5. DIMENSIONS TO BE MET AFTER PLATING. REMOVE ALL BURRS AND SHARP EDGES.

NOTES:

1. THIS STANDARD TAKES PRECEDENCE OVER ANY OTHER DOCUMENTS REFERENCED HEREIN.

2. REFERENCED DOCUMENTS ARE THE ISSUES IN EFFECT AT THE DATE INVITATION FOR BID.

3. THE LIMIT LOAD RATING LISTED CAN BE DEFINED AS THE MAXIMUM RADIAL LOAD WHICH CAN BE APPLIED TO A BEARING WITHOUT IMPAIRING THE SUBSEQUENT FUNCTIONING OF THE BEARING IN AIRFRAME APPLICATIONS. THE ULTIMATE OR STATIC FRACTURE LOAD RATING IS NOT LESS THAN 1.5 TIMES THE LOAD RATING.

PREPARING ACTIVITY: NAVY-AS
CUSTODIANS: ARMY - AT; NAVY - AS; AIR FORCE - 99; DLA -
REVIEW: AF-84 DLA-IS
USER:
PROJECT NUMBER: 3110-0856

MILITARY SPECIFICATION SHEET

TITLE: BEARING, ROLLER, NEEDLE-SINGLE ROW, HEAVY DUTY, SELF-ALIGNING, TYPE III, ANTIFRICTION, INCH

SPECIFICATION SHEET NUMBER: MS24463 — 05 DEC 94 — REV H

SUPERSEDING: MS24463G — 17 JUN 77

AMSC - N/A — FSC 3110

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